

Effects of postharvest time, heat treatment, pH and filtration on the limonin content in Newhall navel orange (*Citrus sinensis* Osbeck cv. Newhall) juice

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Table S1. Effect of postharvest time on the limonin content (mg/L) in juice.

Batch one fruits				Batch two fruits				Batch three fruits	
PTF ^a (d) STJ ^b (d)	9	25	45	0	9	25	45	9	25
0	0.07Da (±0.02)	0.07Da (±0.01)	0.07Ca (±0.02)	0.07Ea (±0.02)	0.06Da (±0.01)	0.06Da (±0.02)	0.06Ba (±0.01)	0.07Da (±0.01)	0.06Da (±0.01)
1	1.86Ca (±0.18)	0.50Cb (±0.08)	0.47Bb (±0.08)	4.62Da (±0.22)	1.59Cb (±0.13)	1.53Cb (±0.08)	0.22Bc (±0.04)	1.33Ca (±0.09)	0.53Cb (±0.06)
3	3.99Ba (±0.19)	1.40Bb (±0.14)	0.87Ac (±0.12)	7.73Ca (±0.49)	5.52Bb (±0.20)	3.31Bc (±0.20)	0.90Ad (±0.09)	2.61Ba (±0.15)	1.40Bb (±0.12)
5	6.38Aa (±0.35)	2.08Ab (±0.18)	1.13Ac (±0.12)	15.72Ba (±0.66)	6.70Ab (±0.40)	4.14Ac (±0.25)	1.02Ad (±0.12)	3.31Aa (±0.18)	1.95Ab (±0.15)
10	6.37Aa (±0.44)	2.07Ab (±0.22)	1.14Ac (±0.16)	24.24Aa (±0.88)	6.68Ab (±0.44)	4.15Ac (±0.25)	1.04Ad (±0.14)	3.34Aa (±0.20)	1.99Ab (±0.17)
15	—	—	—	24.42A (±0.90)	—	—	—	—	—

a: Postharvest Time of Fruits; b: Storage Time of Juices; —: Not applicable; The data were reported as average $\pm$ SD (three replicates); The different lowercase letter in the same line within the same batch of fruits indicated significant difference ($p < 0.05$), the different uppercase letter in the same column within the same batch of fruits indicated significant difference ($p < 0.05$).

Table S2. Effect of heat treatment on the limonin content (mg/L) in juice.

Fruits with 45d of postharvest time				Fruits with 55d of postharvest time		
HTJ ^a						
STJ ^b (d)	15°C	25°C	35°C	25°C	70°C ^c	80°C ^c
0	0.35Da (±0.05)	0.34Da (±0.05)	0.34Ca (±0.04)	0.38Db (±0.07)	2.15Ca (±0.17)	2.15Ca (±0.13)
1	1.49Cc (±0.14)	2.86Cb (±0.11)	3.88Ba (±0.21)	2.34Cb (±0.16)	2.79Ba (±0.18)	2.89Ba (±0.15)
3	2.97Bb (±0.19)	3.82Ba (±0.17)	4.16Ba (±0.25)	3.19Bb (±0.18)	3.52Aab (±0.17)	3.55Aa (±0.17)
5	4.65Aa (±0.20)	4.70Aa (±0.20)	4.70Aa (±0.25)	3.51Aa (±0.21)	3.62Aa (±0.18)	3.65Aa (±0.17)
10	4.67Aa (±0.20)	4.70Aa (±0.18)	4.71Aa (±0.27)	3.60Aa (±0.20)	3.66Aa (±0.19)	3.68Aa (±0.18)

a: Heat Temperatures of Juices; b: Storage Time of Juices; c: The heat treatment lasted only ten minutes, then cooled down and stored at 25°C; The different lowercase letter in the same line within the same postharvest time of fruits indicated significant difference ($p < 0.05$), the different uppercase letter in the same column within the same postharvest time of fruits indicated significant difference ($p < 0.05$). The data were reported as average $\pm$ SD (three replicates).

Table S3. Effects of pH and filtration on the limonin content (mg/L) in juice.

Effect of pH of juice ^a				Effect of filtration of juice ^b		
TMJ ^c STJ ^d (d)	Control	pH=3	pH=10	Control	Filtrated juice	Insoluble tissue (pH=3)
0	0.21Cb (±0.05)	1.98Aa (±0.12)	ND ^e	0.15Ca (±0.02)	0.11Cb (±0.02)	0.03Bc (±0.01)
1	1.64Bb (±0.07)	1.97Aa (±0.11)	ND ^e	1.15Ba (±0.06)	1.05Bb (±0.05)	0.10Ac (±0.02)
3	1.98Aa (±0.08)	1.97Aa (±0.11)	ND ^e	1.28ABa (±0.08)	1.14Ab (±0.05)	0.14Ac (±0.03)
5	2.01Aa (±0.08)	1.98Aa (±0.11)	0.26Bb (±0.03)	1.30Aa (±0.09)	1.15Ab (±0.06)	0.14Ac (±0.03)
10	1.99Aa (±0.08)	1.98Aa (±0.12)	0.41Ab (±0.04)	1.30Aa (±0.10)	1.15Ab (±0.06)	0.14Ac (±0.03)

a: Batch three fruits with 50d of postharvest time were used; b: Batch three fruits with 60d of postharvest time were used; c: Treatment methods of juices; d: Storage time of juices; The different lowercase letter in the same line within the same postharvest time of fruits indicated significant difference ($p < 0.05$), the different uppercase letter in the same column within the same postharvest time of fruits indicated significant difference ($p < 0.05$); e: Not Detected; The data were reported as average $\pm$ SD (three replicates).

